

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460217.1314 +/- 0.0078 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.003 +/- 0.062
Transit depth (Rp/Rs)^2	0.0012 +/- 0.043 [%]
Orbital Inclination (inc)	83.62 +/- 2.82
Airmass coefficient 1 (a1)	0.83 +/- 0.44
Airmass coefficient 2 (a2)	0.1 +/- 0.14
Scatter in the residuals of the lightcurve fit is	51.85 %
Best Comparison Star	None
Optimal Aperture	1.11
Optimal Annulus	3.5
Transit Duration (day)	0.071 +/- 0.072

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.003 ± 0.062 vs 0.1646 (deviation 2.61σ)
Duration fit vs published	0.071 d vs 0.0758 d (deviation 0.07σ)
Mid-time O–C	-2515.2 min
Depth SNR	0.0
Residual scatter	51.85 %

Light curve

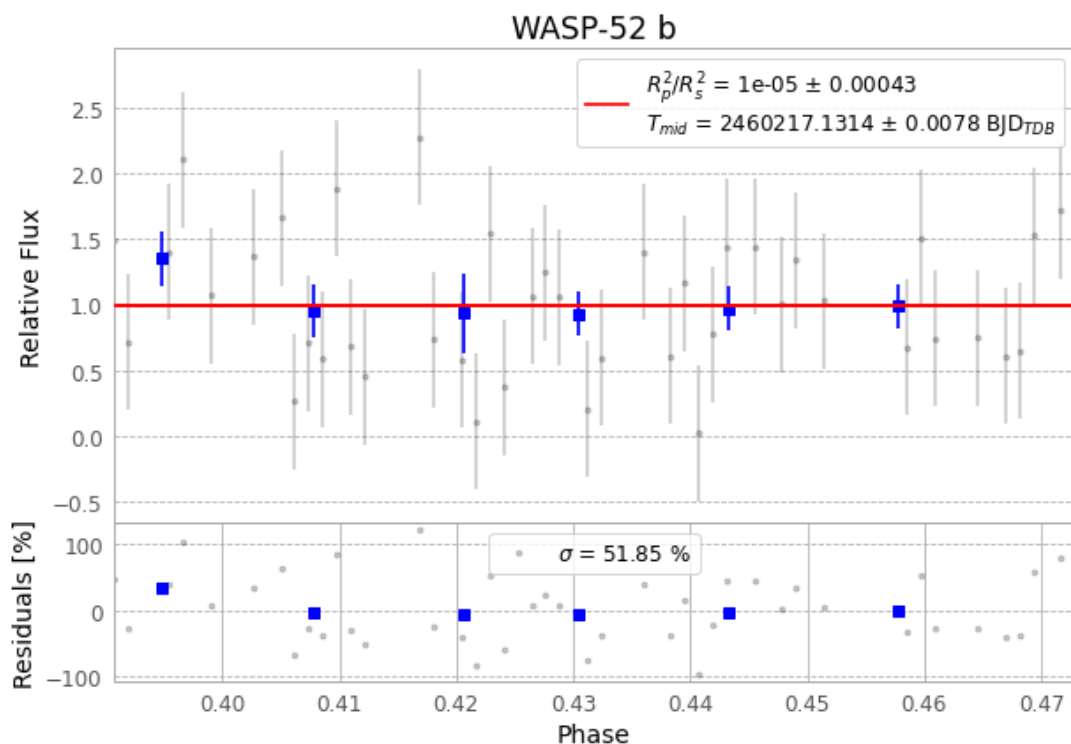


Figure 1 — FinalLightCurve_WASP-52 b_2023-09-30.png (SHA-256 8b98484428184c04...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [211, 85], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4f186210ada4e080...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

70 FITS frames (46 MB), WASP-52230930072415.FITS → WASP-52230930105115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-231001.log (95a3e1d9796e...), FinalLightCurve_WASP-52 b_2023-09-30.png (8b9848442818...), FinalLightCurve_WASP-52 b_2023-09-30.csv (7b8da798f47d...)

Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 16:27Z.